

Customer Churn Prediction

A Machine Learning Risk-Scoring Model for Customer Retention

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Machine Learning Internship Project

Live application: customerchurnprediction.streamlit.app

1. Executive Summary

Customer churn poses a material revenue risk to financial institutions through lost deposits, transaction income, and cross-selling opportunities. This project developed an interpretable churn risk-scoring model using historical customer data from demographics, account activity, digital engagement, and complaint history. Five machine learning algorithms were trained, tuned, and compared; the best-performing model (Logistic Regression, ROC-AUC 0.62) was explained using SHAP to identify the factors driving each prediction, and deployed as a live, interactive tool for real-time risk scoring.

The most important finding is that account tenure is the leading indicator of churn risk in this dataset. Other factors, such as complaint history, appear to be secondary actionable factors. Together with the deployed application, these findings give the institution a practical starting point for prioritizing retention outreach.

2. Introduction

Financial institutions operate in a highly competitive environment where retaining existing customers is generally more sustainable than continuously acquiring new ones. The dataset covered 10,000 customers and 20 variables, including tenure, account type, balances, transaction activity, digital-banking usage, complaints and their resolution, and product holdings. This project uses that data to understand churn behaviour and build a data-driven framework for identifying customers at increased risk of leaving.

3. Objectives

The project set out to answer six strategic questions:

- What is the current customer churn rate and pattern?
- Who is most likely to churn?
- Why do customers churn?
- When are customers most vulnerable to leaving?
- Which customer segments carry higher churn risk?
- How can predictive analytics help identify and retain high-risk customers?

The goal was to develop an interpretable churn prediction and risk-scoring model to support retention strategies that an existing business can use to prevent churn risk.

4. Methodology

4.1 Data Cleaning

The raw dataset (10,000 rows, 20 variables) was checked for duplicates, missing values, and unrealistic observations. No fully duplicate rows were found, though 7,101 rows shared a Customer_ID with unrelated records, an identifier-reuse issue that was documented rather than treated as true duplication. Education_Level was the only column with missing values. This was fixed by imputing an "Unknown" category to avoid biasing any single education group. No impossible values (negative balances, out-of-range rates, invalid churn labels) were found.

4.2 Data Wrangling

The cleaned dataset was filtered to established customers (tenure greater than 12 months, 9,343 rows) to exclude very new accounts that behave differently. Two features were merged: Total_Digital_Usage (combined online and mobile banking activity) and Balance_per_Product (average balance relative to product count). Customers were grouped by account type to compute churn rate, average balance, and engagement activities.

4.3 Exploratory Data Analysis

Eight chart types (histogram, boxplot, scatter plot, bar chart, pie chart, correlation heatmap, pair plot, violin plot) were used to examine distributions and relationships. This stage first surfaced the project's main finding that account tenure showed a strong relationship with churn (correlation ≈ -0.18), while balance, account type, employment status, and digital usage showed little to no relationship.

4.4 Machine Learning

Data was split 80/20 into training and test sets, stratified on churn to preserve the ~26% churn rate in both sets. Five algorithms were trained inside a preprocessing pipeline (standard scaling for numeric features, one-hot encoding for categorical features): Logistic Regression, Decision Tree, Random Forest, XGBoost, and LightGBM. Each was tuned using RandomizedSearchCV with 5-fold stratified cross-validation, optimizing ROC-AUC, an appropriate metric given the class imbalance. The models were evaluated on the held-out test set using Accuracy, Precision, Recall, Specificity, F1-score, ROC-AUC, PR-AUC, and Cohen's Kappa.

4.5 Explainable AI

The selected model (Logistic Regression) was explained using SHAP's LinearExplainer, producing global feature importance, a beeswarm summary plot, Partial Dependence Plots for key features, and individual-prediction waterfall breakdowns for representative high-risk and low-risk customers.

4.6 Deployment

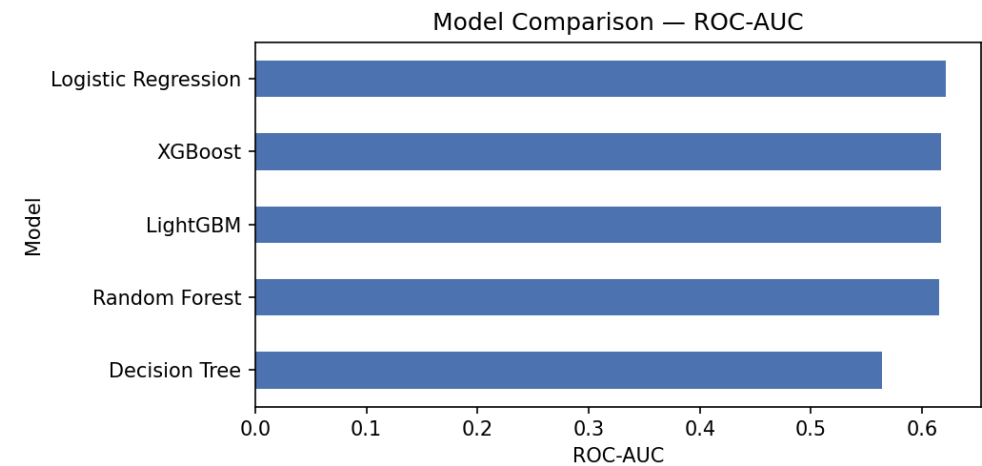
The final model was deployed as an interactive Streamlit application, allowing a user to enter a customer's details and receive a real-time churn probability, a risk category (Low / Medium / High), and a live SHAP-based explanation of the specific factors driving that prediction. The application is hosted publicly at customerchurnprediction.streamlit.app.

5. Results

5.1 Model Comparison

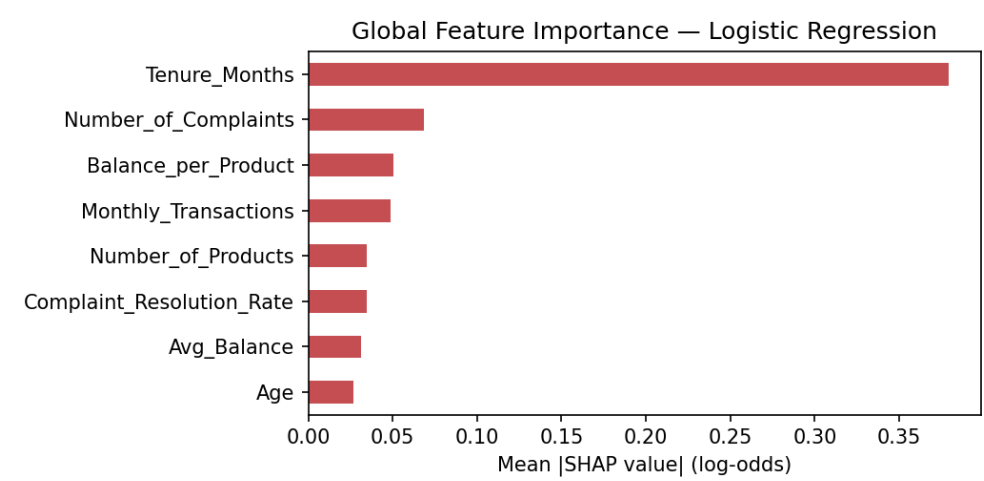
All five tuned models were evaluated on the held-out test set. Logistic Regression achieved the highest ROC-AUC and was selected as the final model.

Model	ROC-AUC	F1-score	Recall	Accuracy
Logistic Regression (selected)	0.622	0.426	0.591	0.579
XGBoost	0.617	0.421	0.569	0.586
LightGBM	0.617	0.417	0.482	0.644
Random Forest	0.616	0.428	0.581	0.590
Decision Tree	0.564	0.405	0.603	0.531



5.2 Feature Importance

SHAP global feature importance confirmed that `Tenure_Months` dominates the model's predictions, contributing roughly 5.5 times more than the next most influential feature, `Number_of_Complaints`. This is consistent with the correlation and visual patterns found during EDA.



6. Interpretation

What is the current churn rate and pattern?

Approximately 26.4% of established customers have churned, a meaningful imbalance that shaped the choice of evaluation metrics in the project.

Who is most likely to churn?

No single demographic condition stands out. Churn rate is nearly unpredictable across gender, employment status, and account type, indicating that demographic profiling alone is not a reliable way to identify at-risk customers.

Why do customers churn?

Shorter account tenure is the strongest available signal, followed by a higher number of complaints. Balance, digital engagement, and product holdings show little independent relationship with churn.

When are customers most vulnerable?

Elevated risk concentrates in the mid-tenure range (roughly 3 to 7 years), rather than immediately after signup, since the analysis already excludes customers in their first year.

Which segments carry higher risk?

Segment-level differences are small: Savings account holders churn marginally more than Fixed Deposit holders, but the gap is not large enough to be a standalone targeting criterion.

How can predictive analytics support retention?

The deployed model and application let the institution score any customer in real time and see the specific factors behind that score, allowing retention resources to be directed toward the customers and factors (particularly complaint resolution speed) that the model and data indicate matter most.

7. Conclusion

This project delivered a complete, interpretable, and deployed churn risk-scoring pipeline. While the model's predictive strength is modest (ROC-AUC ≈ 0.62), reflecting genuinely weak relationships in the available data, the project's greatest value lies in what it rules out as much as what it confirms: churn in this population is not well explained by balance, product count, or demographics, but is meaningfully related to tenure and, to a lesser extent, complaint history. This is a defensible, evidence-based finding that should guide retention strategy more usefully than an overconfident model would.

8. Recommendations

- Prioritize retention outreach for customers in the 3–7-year tenure window, where the data shows elevated risk, rather than treating all tenures equally.
- Invest in faster, more consistent complaint resolution; complaint history is the model's second strongest and most actionable predictor.
- Expand the dataset with richer behavioural signals (e.g., service-call sentiment, channel preference, competitor offers) in future iterations, since current demographic and balance data carry limited predictive signal on their own.
- Retrain the model periodically as new churn outcomes accumulate, to keep the risk scores current as customer behaviour evolves.